

MISSOULA, MT, USA

WMO: 727730

Lat: 46.921N

Lon: 114.093W

Elev: 973

StdP: 90.17

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 24153

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-18.7	-15.4	-23.5	0.5	-18.2	-19.6	0.7	-14.8	10.7	-5.4	9.1	-4.6	1.7	120	0.599

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.2	34.0	16.5	32.2	16.2	30.3	15.7	18.1	29.6	17.3	28.8	16.4	27.8	3.8	10

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.6	11.7	20.3	13.4	10.8	19.9	12.2	9.9	19.2	54.7	29.6	51.9	28.6	49.2	27.8	22.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
9.3	8.2	7.2	DB	-23.4	37.2	4.4	1.5	-26.6	38.2	-29.2	39.1	-31.7	40.0	-34.9	41.1	
			WB	-23.7	19.9	4.4	1.0	-26.8	20.6	-29.4	21.2	-31.8	21.7	-35.0	22.4	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	7.7	-3.5	-1.6	3.2	7.0	11.8	15.8	21.0	19.9	14.5	7.0	0.6	-3.8		
	DBStd	9.53	5.39	5.16	4.11	3.53	3.90	3.90	3.27	3.09	4.05	4.03	4.58	5.20		
	HDD10.0	1917	419	326	212	102	26	3	0	0	9	108	282	429		
	HDD18.3	4073	678	559	469	339	204	96	14	21	124	351	531	688		
	CDD10.0	1082	0	0	1	13	82	177	340	308	145	16	1	0		
	CDD18.3	196	0	0	0	0	2	18	96	69	11	0	0	0		
	CDH23.3	3394	0	0	0	8	97	363	1504	1144	273	5	0	0		
	CDH26.7	1463	0	0	0	1	20	122	726	513	81	1	0	0		
(14) Wind	WSAvg	2.2	1.6	2.0	2.5	2.8	2.7	2.7	2.6	2.4	2.1	1.8	1.7	1.7		
(15) Precipitation	PrecAvg	346	28	21	24	28	44	49	22	26	26	23	25	29		
	PrecMax	554	75	63	53	79	188	107	80	84	91	89	76	113		
	PrecMin	229	4	4	6	2	6	9	0	0	0	0	5	6		
	PrecStd	68	16	14	11	16	29	24	18	21	20	18	17	18		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.0	11.1	18.4	24.3	29.1	33.2	37.1	35.5	31.9	23.6	14.2	8.2		
		MCWB	3.8	5.6	8.5	11.8	14.7	17.1	17.9	16.7	15.7	12.6	9.5	4.8		
	2%	DB	5.7	8.0	14.5	20.7	26.1	30.1	34.8	33.5	29.0	19.5	11.0	5.6		
		MCWB	2.9	4.0	7.0	10.4	13.6	16.0	17.0	16.1	14.7	10.8	6.7	3.1		
	5%	DB	4.1	6.0	12.0	17.6	23.4	27.4	32.9	31.8	26.6	16.4	8.8	3.8		
		MCWB	1.9	2.8	5.9	8.9	12.7	15.0	16.5	15.8	13.9	9.4	5.5	1.8		
	10%	DB	2.7	4.2	9.8	14.5	20.5	25.0	31.0	29.8	24.0	13.7	6.9	2.3		
		MCWB	0.8	1.6	4.7	7.4	11.4	14.0	16.0	15.3	13.0	8.3	4.2	0.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.7	6.4	9.3	12.8	15.8	18.4	20.0	18.7	17.4	13.5	10.0	5.4		
		MCDB	6.9	10.0	16.1	22.8	25.7	29.2	32.2	30.4	26.1	21.4	13.6	7.5		
	2%	WB	3.2	4.6	7.8	10.8	14.5	16.9	18.7	17.5	15.8	11.6	7.2	3.4		
		MCDB	5.2	7.3	13.4	19.2	24.0	27.8	30.3	29.7	26.2	17.9	10.2	5.1		
	5%	WB	2.0	3.1	6.5	9.4	13.3	15.8	17.7	16.7	14.8	10.1	5.8	2.0		
		MCDB	3.9	5.3	11.4	16.4	21.8	25.4	29.4	28.6	24.7	15.2	8.4	3.5		
	10%	WB	1.1	1.9	5.2	8.0	12.0	14.8	16.8	15.8	13.5	8.8	4.6	0.9		
		MCDB	2.4	3.7	9.4	13.9	19.4	23.2	28.4	27.2	22.8	13.1	6.7	1.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	6.8	8.2	10.9	12.4	13.7	14.2	18.2	18.2	15.8	12.0	8.2	6.3		
		MCDBR	8.0	9.9	15.2	18.9	19.5	19.5	21.3	21.6	20.8	16.9	11.1	7.3		
		MCWBR	5.7	6.5	8.7	9.8	8.9	8.0	7.5	7.9	8.9	9.2	7.5	5.5		
		MCWBR	7.7	9.2	14.1	17.2	17.3	17.4	18.6	19.1	18.8	14.9	10.3	6.7		
(40) Clear-Sky Solar Irradiance	taub	taub	0.261	0.273	0.296	0.329	0.339	0.348	0.351	0.384	0.343	0.311	0.284	0.264		
		taud	2.425	2.398	2.444	2.412	2.449	2.454	2.438	2.339	2.434	2.526	2.527	2.458		
		Ebn at Noon	847	909	936	928	925	915	906	860	869	854	817	803		
		Edh at Noon	67	84	95	108	108	109	109	115	96	75	61	58		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	1.25	2.10	3.25	4.59	5.51	6.28	7.29	6.07	4.34	2.60	1.38	0.98		
		RadStd	0.14	0.22	0.30	0.36	0.44	0.53	0.45	0.36	0.44	0.26	0.14	0.10		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	-0.49	N/A	N/A	+55	N/A	
(47) Regional (7 neighbors)		N/A	N/A	N/A	N/A	-0.35	-0.57	N/A	N/A	N/A	N/A	

Nomenclature: See separate page